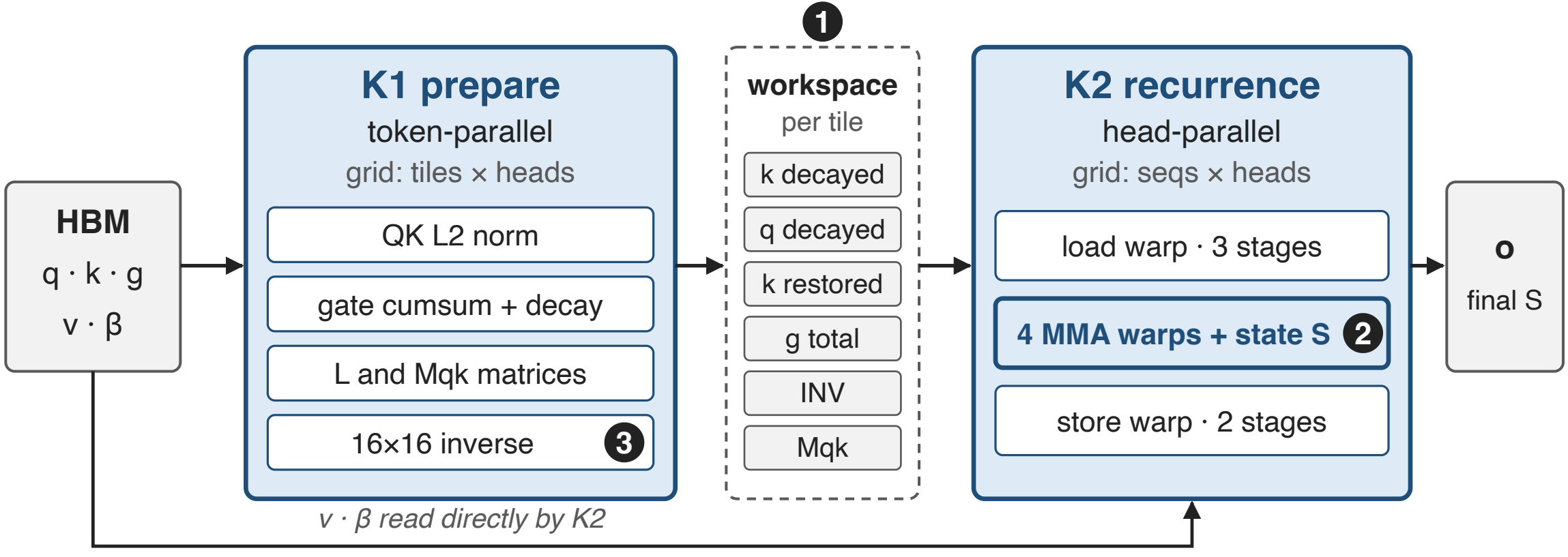

HBM
q · k · g
v · β
K1 prepare
token-parallel
grid: tiles × heads
QK L2 norm
gate cumsum + decay
L and Mqk matrices
16×16 inverse
3
v · β read directly by K2
1
workspace
per tile
k decayed
q decayed
k restored
g total
INV
Mqk
K2 recurrence
head-parallel
grid: seqs × heads
load warp · 3 stages
4 MMA warps + state S
2
store warp · 2 stages
o
final S